

A_∞ Enhancement of Geometry

Koji Sunami

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The goal of our research is to prove homological mirror symmetry conjecture or to develop the relevant area of study. The claim of homological mirror symmetry conjecture is A_∞ -equivalence of Fukaya category $Fuk(X)$ and A_∞ -enhancement $D^b(Coh(X))$ (typically noted as $D^b(X)$) of derived category of coherent sheaves. This statement says the elaboration of mirror symmetry, originally correspondence of symplectic data on Hodge structure and that of complex in Calabi-Yau manifold, whose symplectic side we name as A-model, the complex side as B-model, mirror symmetry solves many of QFT problem as well as Landau-Ginzburg model and others, and homological mirror symmetry is generalization of mirror symmetry in ∞ -category. The proof isn't easy, but the understanding of the problem statement itself is not difficult. I'll write the overview of the problem statement is schematically, precise but not very rigorous, and further I'll write the few possible speculation.

Definition 0.1. A_∞ -Tool Set

(A_∞ -Category)

An A_∞ category $\mathcal{A}$ is a quasi-category which consists of objects $X \in Ob(\mathcal{A})$ and 1-morphisms $f \in Mor^1(\mathcal{A})$, and the 1-morphism contains higher coherence $\alpha : f \Rightarrow g$, called 2-morphism, and iteratively n -morphisms.

(A_∞ -Functor)

For A_∞ category $\mathcal{C}$ and $\mathcal{D}$, A_∞ functor $F : \mathcal{C} \rightarrow \mathcal{D}$ consists of $F_0 : Ob(\mathcal{C}) \rightarrow Ob(\mathcal{D})$ and higher maps $F_n : Hom_{\mathcal{C}}(X_{n-1}, X_n) \otimes \cdots \otimes Hom_{\mathcal{C}}(X_0, X_1) \rightarrow Hom_{\mathcal{D}}(F_0(X_0), F_0(X_n))$. F_0 is a mapping of objects, and $F_1(f)$ is a mapping of morphisms, but it doesn't satisfy the composition $(f \circ g) \circ h \not\cong f \circ (g \circ h)$. Thus the higher map F_n where $n \geq 2$ measures the failure of the composition.

Definition 0.2. Fukaya Category

For a symplectic manifold X , Fukaya category $Fuk(X)$ is an A_∞ category describing Lagrangian intersection. The objects of Fukaya category $Ob(Fuk(X))$ is Lagrangian submanifolds $L \subset X$, and the morphisms are Floer chain complex $CF^*(L_0, L_1)$ for each L_0 and L_1 . This $CF^*(L_0, L_1)$ is a $\mathbb{Z}$ -graded vector space defined by $CF^*(L_0, L_1) = \bigoplus_{p \in L_0 \cap L_1} \mathbb{k}p$. Floer chain complex is generated by its intersection points, and this is equivalent with counting critical points of Morse theory.

In Fukaya category, there is a composition $m_k : CF^*(L_{k-1}, L_k) \otimes \cdots \otimes CF^*(L_1, L_2) \rightarrow CF^*(L_0, L_k)[2-k]$ satisfying Stasheff associahedron property.

A_∞ enhancement of dg-category doesn't have unique enhancement, but the uniqueness is guaranteed if dg-category is derived category $D^b(X)$.

Definition 0.3. A_∞ Derived Category

A_∞ enhancement of derived category $D^b(X)$ is an A_∞ category $\mathcal{A}$, whose homology is $H^*(\mathcal{A}(E, F)) = Ext^*(E, F)$, while its trivialization $H^0(\mathcal{A}(E, F))$ needs to become the based derived category $H^0(\mathcal{A}(E, F)) \cong Hom_{D^b(X)}(E, F)$. This A_∞ enhancement $\mathcal{A}$ can be denoted as $D^b(X)$ by abuse of notation.

Accordingly, homological mirror symmetry conjecture claim that there is an A_∞ equivalence of categories where $D^b(Y)$ is an A_∞ enhanced derived category, and Y is mirror dual of X .

Conjecture 0.1. Homological Mirror Symmetry

The ordinary HMS conjecture is

$$Fuk(X) \cong D^b(Y)$$

Some modern version of HMS conjecture is A_∞ equivalence of dg-category of right hand side

$$Fuk(X) \cong Perf_{dg}(Y)$$

From the above statement of HMS conjecture, the left hand side $Fuk(X)$ describes the symplectic structure called A-model, and right hand side $D^b(X)$ describes complex structure called B-model. The spontaneous question is if there is any different variety of HMS conjecture, possibly more simplified version of that model, or some model already being verified? In particular, original version of mirror symmetry or the QFT version?

The minimum version of mirror symmetry is the correspondence of Hodge structure of symplectic side called A-model and complex side called B-model. According to the outcome of Kähler geometry, symplectic data can be contained in Hodge diamond (why).

Definition 0.4. Mirror Symmetry – Classical Ver

For X the Calabi-Yau 3-fold,

$$h^{1,1}(X) = h^{2,1}(Y)$$

$$h^{2,1}(X) = h^{1,2}(Y) \text{ TBD}$$

The different version of mirror symmetry is QFT. From Marc Gross, classical intersection theory of Kontsevich moduli space whose quantization derives quantum cohomology, which makes semi-infinite variation of Hodge structure in A-model, while period integral of Landau-Ginzburg model generates SCFT, which makes another semi-infinite variation of Hodge structure in B-model. I only introduce A-model side:

Definition 0.5. Mirror Symmetry – QFT Ver

TBD

What is the structure of A_∞ category? The natural question is geometric realization of A_∞ category, which might be Hochschild cochain complex and derive formal scheme, and I use the method of coalgebra construction for them. The geometric interpretation of A_∞ -algebra is noncommutative formal pointed dg-manifold (hereafter in short, ncfp dg-manifold), and A_∞ -category is generalization of A_∞ -algebra. Indeed, similarly A_∞ -category has a correspondence coalgebra, where we can define its geometry.

Definition 0.6. Counital Coalgebra

Coalgebra B is a graded vector space with morphism $\Delta : B \rightarrow B \otimes B$ s.t. $(\Delta \otimes id) \circ \Delta = (id \otimes \Delta) \circ \Delta$ called coproduct, and counital coalgebra is coalgebra B with unit $\epsilon : B \rightarrow 1$ where 1 is the unit object of $\mathcal{C}$, satisfying $(\epsilon \otimes id) \circ \Delta = (id \otimes \epsilon) \circ \Delta$. Counital coalgebra B is cofree counital coalgebra if it's given by $B = C(V) \cong T^c(V)$.

Definition 0.7. Non-Commutative Formal Pointed DG-Manifold

(A Non-Commutative Thin Scheme)

A non-commutative thin scheme X is a functor $X : Alg_{\mathcal{C}^f} \rightarrow Sets$, that is isomorphic to $X \cong Hom(-^*, B_X)$ for some coalgebra B_X , where $\mathcal{C}^f$ is artinian category of finite dimensional objects in $\mathcal{C}$, where $\mathcal{C}$ is $\mathbb{Z}$ -graded vector spaces. In particular, X is a non-commutative formal manifold, if the functor X is isomorphic to some $Spc(T(V))$ for a cofree counital tensor coalgebra $T(V) = \bigoplus_{n \geq 1} V^{\otimes n}$. The dimension of X is defined as $dim_k V$.

(A Non-Commutative Thin DG-Scheme)

A non-commutative thin dg-scheme is a pair (X, d) where X is a non-commutative thin scheme, and d is a vector field on X of degree +1 satisfying bracket of dgla as $[d, d] = 0$. This d is called homological vector field.

(Non-Commutative Formal Pointed DG-Manifold)

A non-commutative formal pointed dg-manifold is a pair $((X, x_0), d_X)$ such that (X, x_0) is a non-commutative formal pointed graded manifold, and d_X is a homological vector field on X such that $d_X|_{x_0} = 0$.

Proposition 0.1. Taylor Decomposition of Graded Vector Field

For the case of formal manifold X , let the corresponding cofree counital coalgebra be $B_X \cong T^c(V)$, and the homological vector field be coderivation $d_X : B_X \rightarrow B_X$, that can have restriction by cofreeness, thus by using projection, $\pi \circ d_X : B_X \rightarrow V[1]$. If in addition, d_X can be decomposed degreewise for its linearity, as $d_X = \sum_{n \geq 1} d_X^{(n)}$ for linear maps $d_X^{(n)} : V^{\otimes n} \rightarrow V[1]$.

Definition 0.8. Non-Unital A_∞ Algebra

A non-unital A_∞ -algebra $A = (A, m_n)$ is a graded vector space A and coherence condition $m_n : A^{\otimes n} \rightarrow A[2 - n]$ with Stasheff relation, whose geometric interpretation is the structure of a nc fp dg-manifold $((X, x_0), d_X)$ together with a coalgebra isomorphism $B_X \cong T(T_{x_0}X)$ with $x_0 : k \rightarrow B_X$.

Definition 0.9. Structure of A_∞ Algebra

The structure of A_∞ Algebra on $V \in Ob(Vect_k^{\mathbb{Z}})$ is given by coderivation d of degree +1 of the non-counital cofree coalgebra $T_+(V[1]) = \bigoplus_{n \geq 1} V^{\otimes n}$ s.t. $[d, d] = 0$.

Proposition 0.2. Stasheff Relation

The condition of nc fp dg-manifold $((X, x_0), d_X)$ derives Stasheff relation of the A_∞ algebra, that is $m_n := d_X^{(n)}$ for $m_n : A^{\otimes n} \rightarrow A$ s.t. $\sum_{r+s+t=n} (-1)^{r+st} m_{r+1+t}(id^{\otimes r} \otimes m_s \otimes id^{\otimes t}) = 0$, whose degree is $|m_n| = 2 - n$. The Taylor decomposition of homological vector field $\pi \circ d_X = \sum_{n \geq 1} d_X^{(n)}$ is degreewise by linearity, as $d_X^{(n)} : B_X^{(n)} \rightarrow B_X^{(1)}$.

Definition 0.10. Non-Commutative Formal n-Pointed DG-Manifold

Nc formal n -pointed dg-manifold $(\mathcal{X}, d_{\mathcal{X}}, \{pt_i\})$ is nc formal graded dg-manifold $(\mathcal{X}, d_{\mathcal{X}})$, and the points are $pt_i \in \mathcal{X}(k)$ for $1 \leq i \leq n$, and coderivation satisfies $d_{\mathcal{X}}|_{pt_i} = 0$, $d_{\mathcal{X}}|_{TX_i} : TX_i \rightarrow \bigoplus_{j,k} TX_j \oplus TX_k$ where X_i is formal neighborhood of X at pt_i .

Definition 0.11. A_∞ -Category

A_∞ -category $\mathcal{A}$ is nc formal dg-manifold $(\mathcal{X}, d_{\mathcal{X}}, \{pt_i\})$ with its coalgebra $B_{\mathcal{X}} \cong T^c(V[1])$. This V has a decomposition $V = \bigoplus_{i,j} E_{ij}$, where $E_{ij} := T(\widehat{X_i \times X_{j_{pt_i, pt_j}}})$.

Remark 0.1. T

is V is a k -linear associative algebra generated from k -linear category $\mathcal{C}$: let $Ob(\mathcal{C}) = I$ to define an algebra $V = \bigoplus Hom(i, j)$ for $i, j \in I$ is a k -linear algebra with the unit $1_V = \sum_{i \in I} id_i$, where $\pi_i = id_i \in V$ is a commuting idempotent i.e. $\pi_i^2 = \pi_i$, $\pi_i \pi_j = \pi_j \pi_i$. This makes homomorphism $k^I \rightarrow V$ of algebra with unit.

Proposition 0.3. Stasheff Relation

From non-unital A_∞ category $\mathcal{A}$, the coalgebra $d_{\mathcal{X}} : T^c(\bigoplus E_{ij}[1]) \rightarrow T^c(\bigoplus E_{ij}[1])$ satisfies the Stasheff relation. Consider the tensor coalgebra can be given by the grading structure $T^c(\bigoplus_{1 \leq i, j \leq n} E_{ij}[1]) = \bigoplus_{m \geq 0} \bigoplus_{1 \leq i_0, i_1, i_2, \dots, i_m \leq n} E_{i_0 i_1} \otimes E_{i_1 i_2} \otimes \dots \otimes E_{i_{m-1} i_m} := \bigoplus_{m \geq 0} B_{\mathcal{X}}^m$. From cofreeness of the coalgebra, the coderivation $d_{\mathcal{X}}$ canonically projected onto $\pi : B_{\mathcal{X}} \rightarrow B_{\mathcal{X}}^1$, and that can be decomposed linearwise, so $\pi \circ d_{\mathcal{X}} = \sum d_{\mathcal{X}}^n$ where $d_{\mathcal{X}}^n : B_{\mathcal{X}}^n \rightarrow B_{\mathcal{X}}^1$, and we have Stasheff relation from $[d_{\mathcal{X}}, d_{\mathcal{X}}] = 0$, and $m_n := d_{\mathcal{X}}^n$.

Remark 0.2

$d_{\mathcal{X}} \circ \eta = 0$ implies $d_{\mathcal{X}} \circ \eta(1) = 0$ or $d_{\mathcal{X}}(1) = 0$. This means curvature is $m_0 = 0$.

Remark 0.3

For the case of $I = \{i\}$ consists of one vertex, we write $d|_{pt} = d \circ \eta$.

For A_∞ category and its relationship with algebraic geometry, Kontsevich-Soibelman claims definition of ncfp dg-manifold of A_∞ algebra and its identification with Hochschild cochain complex as well as formal scheme. This notion can be generalized to A_∞ category including Fukaya category via coalgebra, and I don't know the prospective of future development, but connection of Stable ∞ -category from Lurie might have connection to derived scheme.

On the other hand, there's a transformation of ncfp dg-manifold of A_∞ -algebra into derived formal scheme, and the derived method captures the missing data d_Q used for A_∞ coherence, where ordinary formal scheme cannot have. In dag, we use cdga (A, d) , consisting differential structure on a ring A .

Definition 0.12. Hochschild Cochain Complex

For an associative algebra A , take $A^e = A \otimes A$. Let M as A -bimodule or equivalently A^e -module, we define Hochschild cochain as $CC^n(A, M) = Hom_{A^e}(A^{\otimes n}, M)$, and its differential is given by $d_{Hoch} : CC^n \rightarrow CC^{n+1}$. Its cohomology class is given by homological extension $HC^*(A, M) = Ext_{A^e}^*(A, M)$.

For the case of category $\mathcal{C}$, the Hochschild cochain complex is $HH^*(\mathcal{C}) \cong Ext_{\mathcal{C}^e}^*(\mathcal{C}, \mathcal{C})$ where $\mathcal{C}^e = \mathcal{C}^{op} \otimes \mathcal{C}$.

It should be verified that notion could be well-defined on Fukaya category. Fukaya category has the corresponding coalgebra by Kontsevich-Soibelman, It could be identified with a coderivation complex.

Proposition 0.4. Cobar Construction

$$CC^*(A, A) = CC^*(A) \cong Coder(BA)$$

In particular, $CC^*(Fuk(X)) \cong Coder(BFuk(X))$ where coalgebra of Fukaya category $Fuk(X)$ is $BFuk(X)$.

Considered that in particular shifted Hochschild complex with Gerstenhaber bracket is dgla, I define deformation functor for the study of deformation of dgla. The deformation functor of dgla spontaneously defines formal moduli problem, and I think it will be a wonderful argument, since Fukaya category for example or any other categories has more structure than its Hochschild.

Definition 0.13. Deformation Functor

Let $mathfrak{g}$ be a dgla, and Art is a category of artinian rings. Deformation functor is a functor $Def_{\mathfrak{g}} : Art \rightarrow Set$ s.t.

$$Def_{\mathfrak{g}}(R) := \frac{\{\mathfrak{g} \otimes \mathfrak{m}_R\}}{\text{gauge}}$$

satisfying Schlessinger axiom (TBD).

The application of this Hochschild construction is to derive Gromov-Witten invariants. According to Ganatra, Hochschild cochain complex of Fukaya category derives quantum cohomology by open-closed map, and similarly symplectic cohomology.

Definition 0.14. Quantum Cohomology

TBD

Definition 0.15. Symplectic Cohomology

TBD

Proposition 0.5

$$QH^*(X) \xrightarrow{CO} HH_*(\mathcal{F}(X))$$

$$HH_*(\mathcal{F}(X)) \xrightarrow{OC} QH^{*+n}(X)$$

where CO and OC are closed-open and open-closed map.

So far, Fukaya category has an GW invariant, but GW invariant doesn't describe the whole picture of Fukaya category. As stable ∞ -category has a derived geometric interpretation according to Lurie, I'll consider Fukaya category $Fuk(X)$, and its transformation to stable ∞ -category $Tw^\pi(Fuk(X))$.

Definition 0.16. Stable ∞ -Category

A stable ∞ -category $\mathcal{C}$ consists of zero objects $0 \in \mathcal{C}$ meaning it's object both initial and terminal. $\mathcal{C}$ has all finite limits and finite colimits. A commutative diagram

$$\begin{array}{ccc}
B & \longleftarrow & A \\
\downarrow & & \downarrow \\
D & \longleftarrow & C
\end{array}$$

is a pullback iff it's a pushout.

Though it's different from $Tw^\pi(Fuk(X))$, some example of stable ∞ -category is $\mathcal{C} = Perf(X)$, and the loop space $LX = Map(S^1, X)$ can also be derived self-intersection $LX = X \times_{X \times X}^h X$, connecting to Hochschild complex.

Definition 0.17. Connection to DAG

Meanwhile, I'll consider the formal scheme interpretation without using Hochschild cochain complex.

Definition 0.18. Formal Scheme

A formal scheme is a locally ringed space $(X, \mathcal{O}_X)$ that is locally isomorphic to formal spectrum $Spf(A)$ of a complete topological ring A . Assuming A is I -adically complete for an ideal $I \subset A$ s.t. $A = \varprojlim_n A/I^n$. Then as a set, $|Spf(A)| = \{\mathfrak{p} \subset A \mid \mathfrak{p} \text{ is open and contains } I^n \text{ for some } n\}$. As a remark, prime ideal $\mathfrak{p}$ is defined to be closed in Zariski topology, but from different point of view, it could be open subset in I -adic topology, different from Zariski.

For a scheme X , formal subscheme $\hat{X}_L$ of $L \subset X$ is given by a limit of sheaf $\hat{X}_L = \varprojlim_n \mathcal{O}_X/\mathcal{I}_L^n$ where $\mathcal{I}_L$ is an ideal sheaf of $\mathcal{O}_X$ for L .

For investigating few of the properties, like an ordinary scheme, a formal scheme has closed subschemes in terms of Zariski topology, and any proper close subscheme has codimension at least 1, and in addition the intersection of close subscheme has only finite points (why).

In particular, for the case of Fukaya category $Fuk(X)$, $CF^*(L, L)$ is A_∞ algebra, generating its formal scheme $Spf(Sym(CF^*(L, L)^\vee[-1]))$. The differential structure d_Q was a direct consequence of coderivation $Q : T^c(CF^*(L, L)^\vee[-1]) \rightarrow T^c(CF^*(L, L)^\vee[-1])$, with A_∞ coherence.

Example 0.1. Non-Commutative Formal Pointed DG-Manifold

For coalgebra derivation $Q : T^c(V[1]) \rightarrow T^c(V[1])$ s.t. $Q^2 = 0$ of A_∞ structure where V is a vector space, there is a correspondence to formal scheme $(V, Q) \leftrightarrow (Spf(A), d_Q)$, where $V^\vee$ is a dual vector space, and $A = Sym(\hat{V}^\vee)$, and $d_Q : Spf(A) \rightarrow Spf(A)$ is a differential s.t. $d_Q(f) = Q(f)$ for every function $f \in A$.

So far ncfp dg-manifold has interpretation of formal geometry, and it was a domain of algebraic geometry. However, ncfp dg-manifold contains indispensable ingredient d_Q other than $Spf(A)$, not atually defined in formal scheme yet, but derived formal scheme includes differential d_Q data in itself. For defining derived formal scheme:

Definition 0.19. Derived Formal Scheme

Derived formal scheme is a functor $cdga \rightarrow \infty\text{-groupoid}$ s.t. $\pi_0(A) = Spf(A)$.

As an example, for the case of Fukaya category $Fuk(X)$, $CF^*(L, L)$ is A_∞ algebra, and it generates $Spf(Sym(CF^*(L, L)^\vee[-1]))$. Adding the derived structure, it will be (TBD).

So far we created derived formal scheme from Fukaya category in two ways: one uses Hochschild and the

other is not. What is the relationship between them if any? So to summarize the above arguments, studying Fukaya category involves:

- Quantum/Symplectic cohomology
- Stable ∞ -category
- Derived formal scheme

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1 Deformation Theory

Deformation theory is literally a study of deformation functor, but the contextual meaning is to track infinitesimal change of geometry.

I don't know if there's a unified way of formalizing deformation theory, but one thing is solution space of Maurer-Cartan equation by gauge equivalence, that's noted by deformation functor.

2 Stable ∞ -Category

3 Balmer Spectrum

If A_∞ category is symmetric monoidal category, its homotopy category is tensor triangulated category. In particular, tensor stable ∞ -category's homotopy category is tensor triangulated category.

Definition 3.1. Tensor Triangulated Category

Triangulated category with monoidal structure.

Balmer spectrum is a Zariski spectrum induced from tensor triangulated structure to define the geometric realization of triangulated category. A notable non-obvious example of tensor triangulated category is stable homotopy category.

Definition 3.2. Balmer Spectrum

Definition 3.3. Stable Homotopy Theory

4 Quantization and Deformation

I don't know where to put this section to the introduction, and it might be a branch of main question of this pdf. Away from category, I go back to the basic of abstract algebra to discuss examples of associativity, in particular quantization and deformation.

Recall non-commutative scheme from its coalgebra has an identification of A_∞ algebra/category. Consider a functor $X : Alg \rightarrow Coalg$, mapping an associative algebra $O(X)$ to a coalgebra B_X . Associative algebra is more general than unital commutative associative algebra $CRing$, and it's different from classical commutative scheme. Some idea of understanding noncommutative scheme is

Recall a commutative ring defines an affine scheme by its ring spectrum $X = V(\mathfrak{p})$ with its structure sheaf $\mathcal{O}_X$, and that can also be described functorially as $F : CRing \rightarrow Set$. Hereafter I focus on functorial point of view.

- Deformation
- Quantization

I don't know the general classification of associative algebra, but one important fact is that commutative algebra's quantization is associative. I wonder if there is any connection to the classical algebraic geometry or derived algebraic geometry in general.

Remark 4.1. Associative Algebra Example

To name some examples of associative algebra, $U(\mathfrak{g})$, $k[G]$, $M_n(k)$, Hecke algebra, path algebra, quiver algebra etc, and quantization of associative algebra is associative algebra.

The spontaneous question is what is the invariant under the quantization operation? Notice especially the boring part is the matrix ring $M_n(k)$, since it's deformation invariant so it's Morita equivalent to k , but I'd also wonder the quantized version.

Definition 4.1. Morita Equivalence of Quantization

TBD

Question 4.1

Should we just use deformation of associative algebra, which makes A_∞ algebra?

Another approach is deformation (TBD).

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